

Gaussian kernels fail on every positive-dimensional closed Riemannian manifold

Candidate full result for arXiv:2303.06558

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Status. Candidate full result, likely valid. We prove the qualitative extension suggested in the source: for every positive-dimensional closed Riemannian manifold, including every simply connected one, the intrinsic Gaussian kernel is not positive definite at any positive bandwidth.

1 Source question and result

For a metric space (X, d) , write

$$\Lambda_+(X) = \{\lambda > 0 : e^{-\lambda d(\cdot, \cdot)^2} \text{ is positive definite}\}.$$

Siran Li's source paper [1] proves that $\Lambda_+(M, d_g) = \emptyset$ for every non-simply-connected closed Riemannian manifold. It records the suggestion that the same conclusion should hold for *every* closed Riemannian manifold, but says that it cannot confirm the suggestion or provide a counterexample. The exact passage is reproduced in Figure 1.

Proposition 1.6. *On any complete Riemannian manifold $(\mathcal{M}, g)$ and for any $q > 2$, there is some $\lambda > 0$ such that the kernel $k(x, y) := e^{-\lambda[d_g(x, y)]^q}$ fails to be positive definite.*

1.3. **Further comments.** It is remarked in [9] that $\Lambda_+(\mathbf{S}^1) = \emptyset$ — hence Theorem 1.4 — can be deduced via alternative arguments from Wood [31] and Gneiting [14]. Nevertheless, the hands-on Fourier theoretic computations in [9] are particularly suitable for extensions to more general classes of Riemannian manifolds, as will be shown in this contribution.

On the other hand, [9] suggests that one may deduce $\Lambda_+(\mathcal{M}, g) = \emptyset$ for any closed Riemannian manifold, by showing that any such $(\mathcal{M}, g)$ admits an isometric embedding of the round circle $\mathbf{S}^1$ in the metric space sense. This, in fact, seems to impose rather strong limitations on $(\mathcal{M}, g)$. See the theorem in Appendix A. But at the moment we are unable to provide a counterexample, nor to confirm the suggestion in [9].

Let us also remark that variants of Gaussian kernels on $\mathbf{S}^n$ and more general Riemannian manifolds are studied in theoretical works in mathematics, statistics, and machine learning. See [3, 11, 12, 14] and many others.

2. NON-POSITIVE DEFINITENESS OF GAUSSIAN KERNEL WHEN $\pi_1(\mathcal{M}) \neq \{0\}$

In this section we prove our main theorem, which is stated in §0 and reproduced below:

Theorem 2.1. *Let $\mathcal{M}$ be an arbitrary closed (i.e., compact and boundaryless) differentiable*

Figure 1: The unresolved all-closed-manifolds suggestion on page 3 of the official arXiv PDF.

Transcription of the source statement. “On the other hand, [9] suggests that one may deduce $\Lambda_+(M, g) = \emptyset$ for any closed Riemannian manifold, by showing that any such (M, g) admits an isometric embedding of the round circle S^1 in the metric space sense. This, in fact, seems to impose rather strong limitations on (M, g) . See the theorem in Appendix A. But at the moment we are unable to provide a counterexample, nor to confirm the suggestion in [9].”

Idea of the proof

The source’s obstruction uses a closed geodesic which is almost globally distance minimizing. Such control is unavailable in the simply connected case. The proof below instead uses only local minimality and the rigidity of analytic positive-definite feature maps. Along a short geodesic segment the feature curve has exactly the line-Gaussian Gram matrix, which forces it to be real analytic. On a closed geodesic, analytic continuation is incompatible with periodicity.

Theorem 1 (full qualitative resolution). *Let (M, g) be a positive-dimensional closed Riemannian manifold. For every $\lambda > 0$, the kernel*

$$k_\lambda(x, y) = \exp(-\lambda d_g(x, y)^2)$$

is not positive definite. Equivalently,

$$\boxed{\Lambda_+(M, d_g) = \emptyset.}$$

We prove a metric obstruction which makes the mechanism transparent.

2 The line-Gaussian feature curve

Fix $\lambda > 0$. Define $\psi_\lambda : \mathbb{R} \rightarrow \ell^2(\mathbb{N}_0; \mathbb{R})$ by

$$(\psi_\lambda(t))_n = e^{-\lambda t^2} \frac{(2\lambda)^{n/2} t^n}{\sqrt{n!}}, \quad n \geq 0. \quad (1)$$

The exponential series gives

$$\langle \psi_\lambda(s), \psi_\lambda(t) \rangle = e^{-\lambda(s^2+t^2)} e^{2\lambda st} = e^{-\lambda(s-t)^2}. \quad (2)$$

This curve is real analytic in Hilbert norm. More explicitly, for $z \in \mathbb{C}$ the same formula defines a vector in $\ell^2(\mathbb{N}_0; \mathbb{C})$ and

$$\sum_{n \geq 0} \left| e^{-\lambda z^2} \frac{(2\lambda)^{n/2} z^n}{\sqrt{n!}} \right|^2 = e^{-2\lambda \operatorname{Re}(z^2) + 2\lambda |z|^2} = e^{4\lambda (\operatorname{Im} z)^2}. \quad (3)$$

Thus the series converges locally uniformly in Hilbert norm and is an entire $\ell^2(\mathbb{C})$ -valued map.

Lemma 2 (local Gram rigidity). *Let $I \subset \mathbb{R}$ be an open interval and $F : I \rightarrow H$ a map into a real Hilbert space such that*

$$\langle F(s), F(t) \rangle_H = e^{-\lambda(s-t)^2} \quad (s, t \in I).$$

Then F is real analytic in Hilbert norm.

Proof. On the algebraic span of $\{\psi_\lambda(t) : t \in I\}$ define

$$U \left(\sum_j a_j \psi_\lambda(t_j) \right) = \sum_j a_j F(t_j).$$

The asserted equality of Gram matrices shows that the two sides always have the same squared norm. Hence U is well defined and isometric, and it extends to the closed span. In particular, $F(t) = U\psi_\lambda(t)$ on I . Equations (1)–(3) now prove Hilbert-norm real analyticity. $\square$

3 Periodic local geodesics are impossible

Call a map $\gamma : \mathbb{R} \rightarrow (X, d)$ a *periodic local unit geodesic* if it has a period $L > 0$ and, for every $a \in \mathbb{R}$, there is an open interval I_a containing a such that

$$d(\gamma(s), \gamma(t)) = |s - t| \quad (s, t \in I_a). \quad (4)$$

Lemma 3 (periodic-geodesic obstruction). *If a metric space contains a periodic local unit geodesic, then $e^{-\lambda d^2}$ is not positive definite for any $\lambda > 0$.*

Proof. Suppose the kernel is positive definite. By the standard feature-map construction there are a real Hilbert space H and a map $\Phi : X \rightarrow H$ with

$$\langle \Phi(x), \Phi(y) \rangle_H = e^{-\lambda d(x,y)^2}. \quad (5)$$

Set $F = \Phi \circ \gamma$. On every interval I_a , equations (4) and (5) give the hypothesis of Lemma 2. Therefore $F : \mathbb{R} \rightarrow H$ is locally, hence globally, real analytic.

Fix $a \in \mathbb{R}$. The scalar function

$$h_a(t) = \langle F(a), F(t) \rangle_H$$

is real analytic on the connected line. On a neighborhood of a , local minimality gives

$$h_a(t) = e^{-\lambda(t-a)^2}.$$

The identity theorem for real-analytic functions forces this equality for every $t \in \mathbb{R}$. But $\gamma(a+L) = \gamma(a)$, so (5) yields

$$1 = \|F(a)\|^2 = h_a(a+L) = e^{-\lambda L^2} < 1,$$

a contradiction. □

4 Application to closed manifolds

Proof of Theorem 1. The Lyusternik–Fet theorem [2] states that every positive-dimensional closed Riemannian manifold has a nonconstant closed geodesic. Parametrize one by unit speed and lift its periodic parameter to $\gamma : \mathbb{R} \rightarrow M$, with period equal to its positive length. Every Riemannian geodesic is distance minimizing on sufficiently short parameter intervals. Thus γ satisfies (4), and Lemma 3 excludes positive definiteness for the arbitrary parameter $\lambda > 0$. □

The same proof applies to any metric space with a periodic local unit geodesic; neither a globally isometrically embedded round circle nor a globally shortest closed geodesic is needed. This is exactly why the argument crosses the simple-connectivity barrier identified in the source.

5 Verification, novelty, and boundary

The included verifier checks finite truncations of (1) against the exact Gram identity (2) and evaluates the strictly positive separation

$$\|\psi_\lambda(0) - \psi_\lambda(L)\|^2 = 2(1 - e^{-\lambda L^2}) > 0,$$

which periodicity would force to vanish. The mathematical audit in `VERIFIER_REPORT.md` separately checks feature-map existence, the local Gram isometry, Hilbert-valued analyticity, scalar continuation, and the Lyusternik–Fet reduction.

Bounded searches of the run indexes, the exact source suggestion, later citations, and combinations of “Gaussian kernel”, “closed geodesic”, and “analytic continuation” through 13 August 2026 found no later proof. A 2025 experimental study still presents the geometry-versus-bandwidth problem as open. Novelty confidence is high enough for promotion but remains subject to expert literature review, especially for prior general theorems propagating near-diagonal analyticity of positive-definite kernels.

Dimension zero is essential: as the source notes, finite metric spaces can have a nonempty proper positivity range. The result settles the qualitative closed-manifold conclusion. It does not quantify how many sample points, or what spacing, is needed to exhibit a negative Gram eigenvalue at a fixed λ ; that finite-scale problem remains open.

Human-review recommendation. Audit Lemma 2’s Gram-isometry construction and the use of the real-analytic identity theorem in Lemma 3. Then perform a specialist search in the regularity theory of positive-definite kernels before asserting priority.

References

- [1] S. Li, *Gaussian kernels on non-simply-connected closed Riemannian manifolds are never positive definite*, Bull. Lond. Math. Soc. **56** (2024), 263–273; arXiv:2303.06558.
- [2] L. A. Lyusternik and A. I. Fet, *Variational problems on closed manifolds*, Dokl. Akad. Nauk SSSR (N.S.) **81** (1951), 17–18 (in Russian).